

Gustavo Gonzalez Ramos

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EDUCATION

Tecnológico de Monterrey (ITESM)

Expected Graduation: [May 2028]

School of Computer Science and Information Technologies

Engineering in Computer Science and Technology

Average Grade: 94/100

Relevant Coursework:

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, HTML5, CSS

Technologies and Frameworks: React, NextJs, NestJs, Tailwind CSS, Node.js, Astro, Vite, Express.js, Docker, PostgreSQL, Redis, Prisma, Supabase

Software and Tools: Git, GitHub, Visual Studio Code, Linux, Postman

Extras: Terminal, Markdown, Bash, npm, pnpm

Languages: Spanish (Native), English (Advanced - C1)

WORK EXPERIENCE

Coppel

February 2024 – May 2024

Intern – Software Engineer (Full Stack Developer)

(Unpaid internship)

- Improved and optimized PostgreSQL functions to enhance database performance and data retrieval efficiency.
- Developed and maintained features for the **Sistema Integral de Exhibición (SIE)**, a platform managing product displays across multiple stores.
- Built and integrated a full-stack web application with **Angular and TypeScript** for the front-end and **PHP** for the backend.
- Deployed and managed cloud-based services on **Microsoft Azure**, ensuring scalability, security, and performance.
- Collaborated with cross-functional teams to implement new system features and optimize existing workflows.

Skills: PostgreSQL, Angular, TypeScript, PHP, Azure

PROJECTS

BiciTec

Solo Project — Engineering Competition

IoT-Web Bicycle Sharing Management Platform for Campus

- Designed and built an IoT-enabled system to control and monitor the institutional bicycle service at ITESM Campus Monterrey.
- Integrated sensors on stations and bicycles to automatically record loans, returns and speed data in real time.
- Developed a full web platform with separate dashboards for students and administrators, including bilingual support (Spanish and English).
- Implemented secure authentication and authorization using **JWT**, institutional ID and role-based access (user/admin).
- Designed a relational schema in **MySQL** to store users, bikes, stations and trip records, enabling full traceability of mobility patterns.
- Used **Redis** for session management, token caching and overall backend performance improvement.
- Built the frontend with reusable **Next.js** and React components, ensuring a responsive and consistent UI for real-time supervision.

- Developed a REST API with **Node.js** and **Express** to handle trips, speed monitoring, user registration and communication with IoT sensors.
- Deployed the platform on **Railway**, integrating backend, frontend, database and IoT services into a unified cloud environment.

Technologies: Next.js, React, TypeScript, Tailwind CSS, Node.js, Express, MySQL, Redis, JWT, IoT Sensors, Railway

FitTrack

Solo Project

Fitness Tracking and Progress Monitoring Application

- Developed a fitness tracking app to record and monitor users workout sessions, including cardio and strength training.
- Implemented a real-time tracking system for the user's progress with detailed logs for each workout.
- Designed a user-friendly interface for tracking workout data such as distance, duration and session types.
- Integrated a feature to track historical progress with data visualization (graphs and charts).
- Built reusable React components, ensuring consistency and maintainability throughout the app.
- Leveraged **TypeScript** for type safety and **Vite** for fast development and building.
- Developed a backend API using **Node.js** and **Express**, with PostgreSQL as the database for secure data storage.
- Implemented user authentication and authorization with **JWT** and managed sessions with secure cookies.
- Utilized **Zod** for runtime validation of data structures and ensuring type safety on both client and server sides.

Technologies: React, TypeScript, Vite, CSS Modules, Node.js, Express, PostgreSQL, JWT, Cookies, Zod

ChefOps

Solo Project

Real-Time Kitchen Order Management Application for Restaurants

- Developed a kitchen order management system to track orders by status (Queued, Cooking, Completed, Canceled) in real-time.
- Implemented real-time UI features, including a digital clock for an enhanced user experience.
- Created modular components and services, ensuring a scalable, maintainable codebase.
- Designed and developed reusable UI components such as **Clock**, **MenuTabs**, and **NewOrder-Button**.
- Built custom React hooks (**useAllOrders**, **useOrder**) for efficient state management and order data handling.
- Leveraged **TypeScript** and **Vite** for enhanced performance and type safety.
- Ensured scalability and fast performance with a component-based architecture and efficient service management.

Technologies: React, TypeScript, Vite, Tailwind CSS